Computer Vision — Image Classification with Transfer Learning

DAC-204 Programming for DS and AI
Lab Assignment: Image Classification using Transfer Learning

Total Marks: 10

Problem

Image classification is a fundamental task in computer vision where a machine learning model identifies the category of an image. Training deep learning models from scratch requires large datasets and significant computational power.

Transfer learning solves this problem by using **pretrained CNN models such as VGG16, ResNet, or MobileNet** that already learned features from large datasets like ImageNet.

In this lab, you will use a pretrained convolutional neural network to classify images and analyze how transfer learning improves performance.

Dataset

Dataset: CIFAR-10 Image Dataset

Link: **https://www.kaggle.com/datasets/ayush1220/cifar10**

Contains **60,000 images in 10 classes**

Classes include:

- airplane
- automobile
- bird
- cat
- deer
- dog
- frog
- horse
- ship
- truck

You can take reference from this paper: Learning Multiple Layers of Features from Tiny Images

Author: Alex Krizhevsky

Objectives

• Understand image datasets and preprocessing
• Apply transfer learning using pretrained CNN models
• Train and evaluate an image classification model

• Compare transfer learning vs training from scratch
• Interpret model performance using confusion matrix

1. Data Understanding & Visualization (2 marks)

i) Load Dataset

• Load CIFAR-10 dataset using TensorFlow/Keras
• Print dataset shape
• Display sample images with labels

ii) Data Exploration

• Check number of classes
• Count images per class
• Visualize at least **10 random images**

2. Image Preprocessing (2 marks)

i) Normalization

• Normalize pixel values between **0 and 1**

ii) Train-Test Split

• Use standard dataset split

iii) Data Augmentation

Apply at least two techniques:

• Rotation
• Horizontal Flip
• Zoom

Use **ImageDataGenerator**

3. Model Building (2 marks)

i) Baseline Model

• Train a simple CNN model with:

- Conv layer
- MaxPooling
- Dense layers

ii) Transfer Learning Model

Use a pretrained network such as:

• MobileNetV2
or
• VGG16

Steps:

- Load pretrained model
- Freeze convolution layers
- Add custom classification layer

4. Model Evaluation (2 marks)

Evaluate both models using:

- Accuracy
- Confusion Matrix
- Classification Report

Compare:

- Baseline CNN vs Transfer Learning

5. Interpretation & Reflection (2 marks)

i) Feature Learning

Explain how pretrained CNNs help extract useful image features.

ii) Reflection Questions

1. Why is transfer learning useful for small datasets?
2. What happens if pretrained layers are not frozen?
3. When can transfer learning perform worse than training from scratch?